Secure Sight Statistical Analysis Report

Total runs: 25

Total batches: 5

1. Global Metrics

Metric	Mean	Std Dev	Variance	Min	Max	Median	Count
accuracy	0.882542	0.0193645576035860 6	0.000360	0.848468	0.915821	0.884026	25
precision	0.872027	0.0255890782745505 36	0.000629	0.827591	0.928699	0.876727	25
recall	0.883841	0.0255729570302154 8	0.000628	0.837560	0.929544	0.886608	25
f1_score	0.875067	0.0205883343522711 84	0.000407	0.836546	0.914991	0.873292	25
auc	0.895226	0.0278325949631116 1	0.000744	0.842868	0.948310	0.899564	25
latency	9.411680	1.4892998067497136	2.129293	7.755605	12.088238	9.603186	25

2. Confidence Intervals

Metric	Mean	CI Lower	CI Upper
accuracy	0.882542	0.874548	0.890535
precision	0.872027	0.861464	0.882590
recall	0.883841	0.873285	0.894397
f1_score	0.875067	0.866569	0.883566
auc	0.895226	0.883737	0.906715
latency	9.411680	8.796927	10.026433

3. Per-Batch Metrics

Batch 1

Metric	Mean	Std Dev	Variance	Min	Max	Median
accuracy	0.886268	0.0119444215754667	0.000114	0.868468	0.895821	0.891550
precision	0.872166	0.0246193157340565 23	0.000485	0.847591	0.908699	0.876727

recall	0.892712	0.021994546 07605115	0.000387	0.857560	0.909544	0.90156 9
f1_score	0.882289	0.013101402 848217064	0.000137	0.860718	0.892784	0.88321 6
auc	0.898001	0.018620549 397197966	0.000277	0.868274	0.911645	0.90956 4
latency	7.942580	0.124279079 4006885	0.012356	7.755605	8.069188	8.00265 5

Batch 2

Metric	Mean	Std Dev	Variance	Min	Max	Median
accuracy	0.860797	0.012714226 039313771	0.000129	0.848468	0.875821	0.85967 8
precision	0.847039	0.026059802 354930654	0.000543	0.827591	0.888699	0.83108 8
recall	0.862315	0.024222705 933271325	0.000469	0.837560	0.889544	0.86534 4
f1_score	0.855875	0.014457348 164581853	0.000167	0.840718	0.872784	0.86194 0
auc	0.869743	0.021188960 676254497	0.000359	0.848274	0.891645	0.87095 7
latency	7.955887	0.135384970 29374835	0.014663	7.755605	8.069188	8.00265 5

Batch 3

Metric	Mean	Std Dev	Variance	Min	Max	Median
accuracy	0.902635	0.016385511 72395041	0.000215	0.881516	0.915821	0.91155 0
precision	0.894962	0.021758254 725068156	0.000379	0.871088	0.928699	0.89672 7
recall	0.902107	0.025812732 31739681	0.000533	0.872475	0.929544	0.90938 6
f1_score	0.895396	0.021076724 004960655	0.000355	0.866546	0.914991	0.90194 0

auc	0.914132	0.031954303662711785	0.000817	0.872868	0.948310	0.929564
latency	9.634360	0.20516477632479757	0.033674	9.306726	9.858685	9.683026

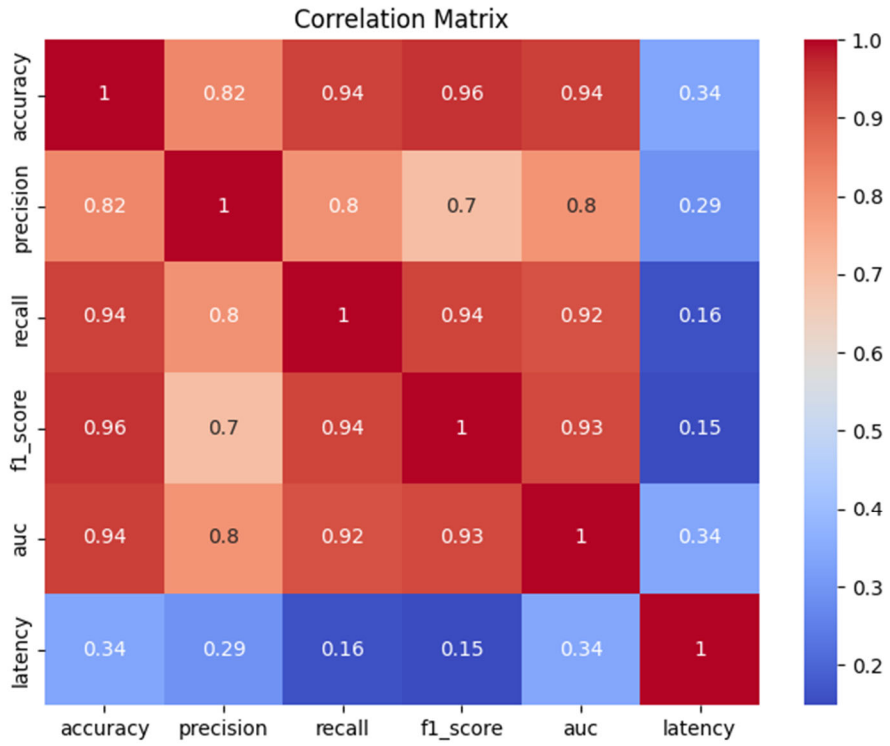
Batch 4

Metric	Mean	Std Dev	Variance	Min	Max	Median
accuracy	0.872635	0.016385511 72395041	0.000215	0.851516	0.885821	0.881550
precision	0.864962	0.021758254 72506816	0.000379	0.841088	0.898699	0.866727
recall	0.872107	0.025812732 31739681	0.000533	0.842475	0.899544	0.879386
f1_score	0.865396	0.021076724 004960655	0.000355	0.836546	0.884991	0.871940
auc	0.884132	0.031954303 662711785	0.000817	0.842868	0.918310	0.899564
latency	9.634360	0.20516477632479757	0.033674	9.306726	9.858685	9.683026

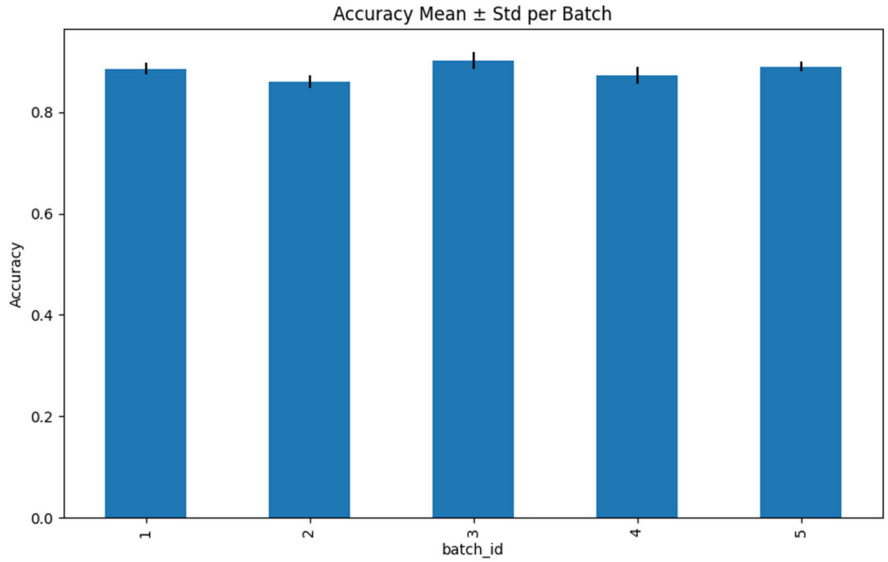
Batch 5

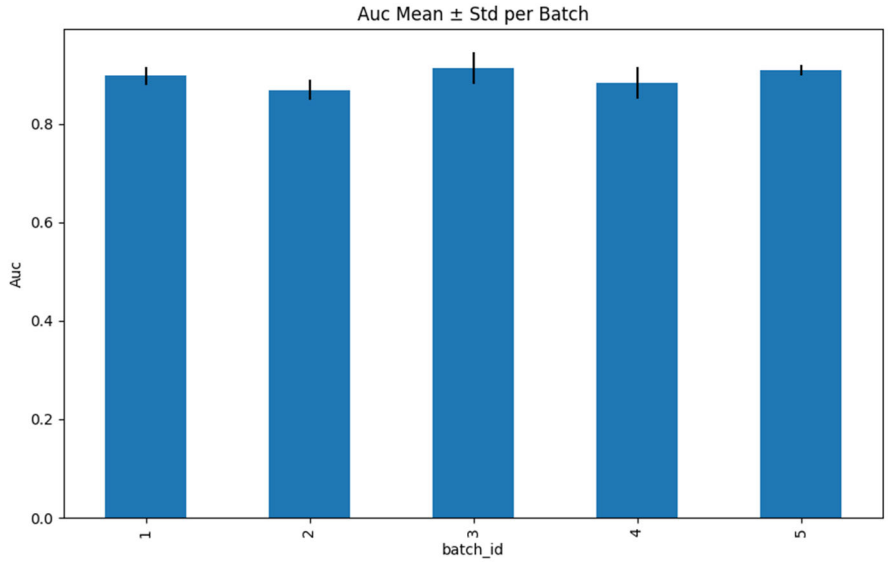
Metric	Mean	Std Dev	Variance	Min	Max	Median
accuracy	0.890374	0.009808132 162759992	0.000077	0.881704	0.901695	0.884026
precision	0.881007	0.009771394 653209288	0.000076	0.864985	0.889313	0.881116
Recall	0.889965	0.014392742 289228875	0.000166	0.869431	0.906155	0.886608
f1_score	0.876381	0.010626368 899567607	0.000090	0.862733	0.891283	0.873292
Auc	0.910123	0.010519467 412790997	0.000089	0.892192	0.919748	0.911934
Latency	11.891212	0.18460271633512396	0.027263	11.694424	12.088238	11.970417

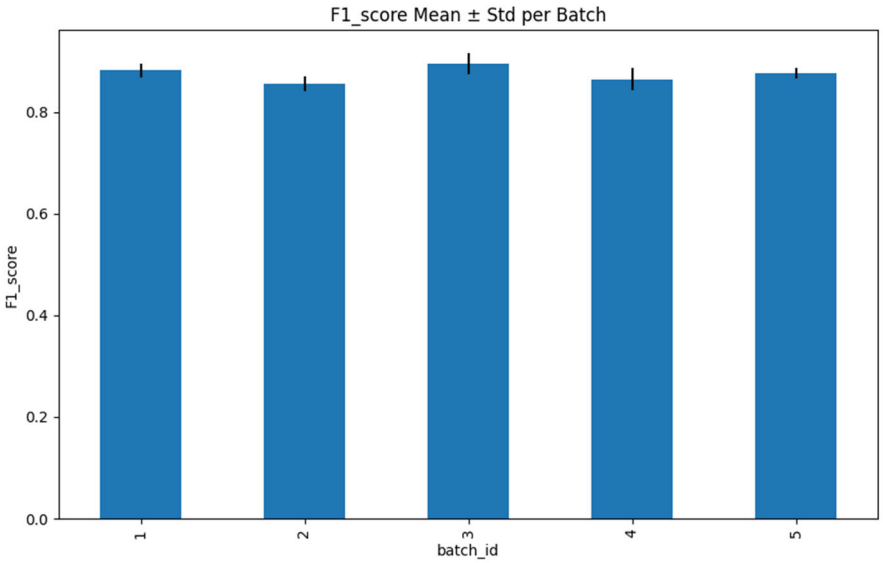
4. Correlation Analysis

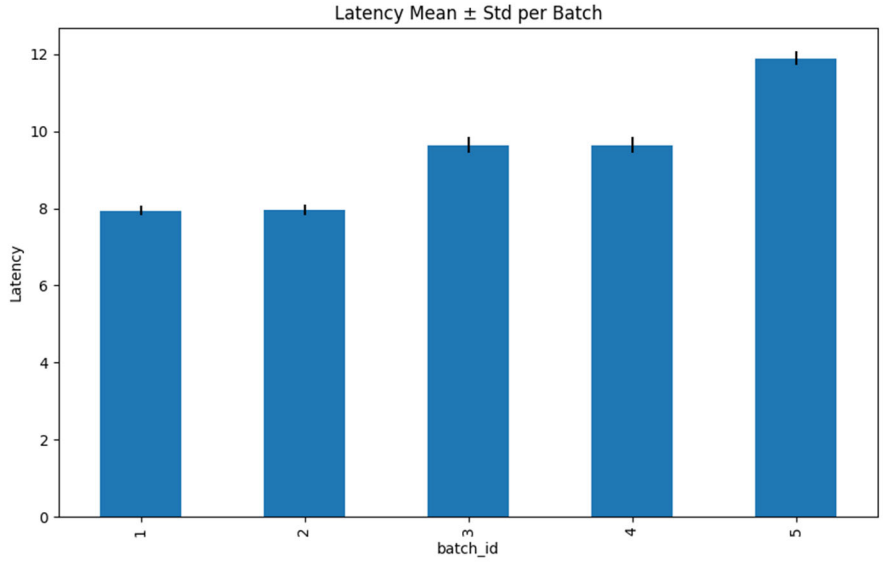


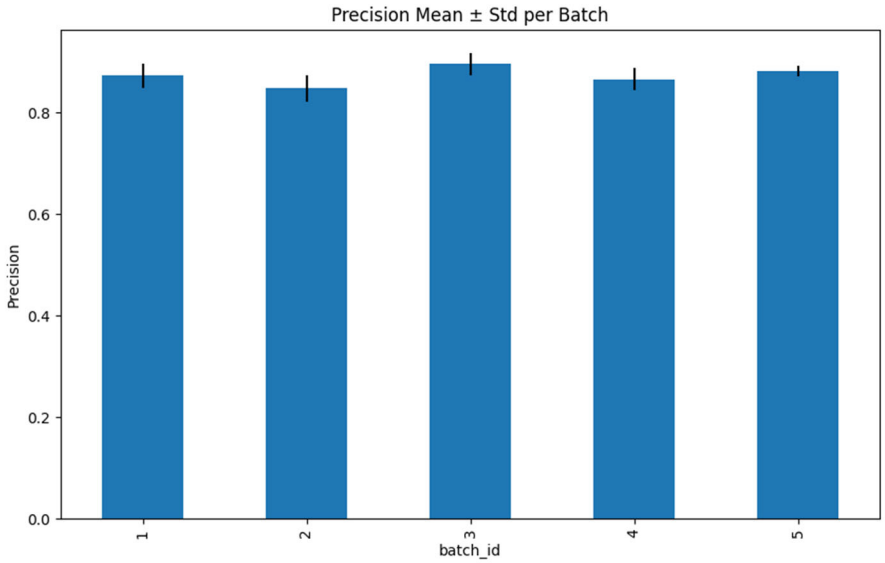
5. Visualizations

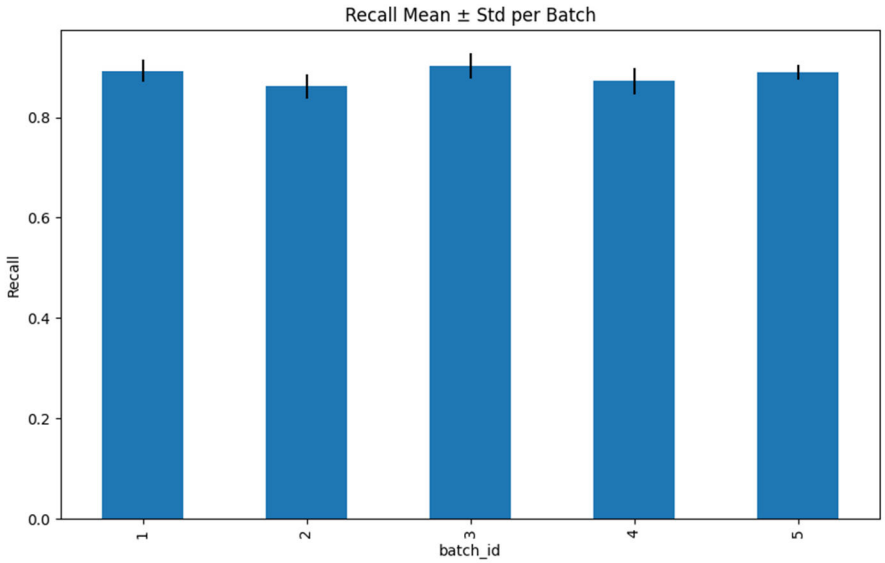


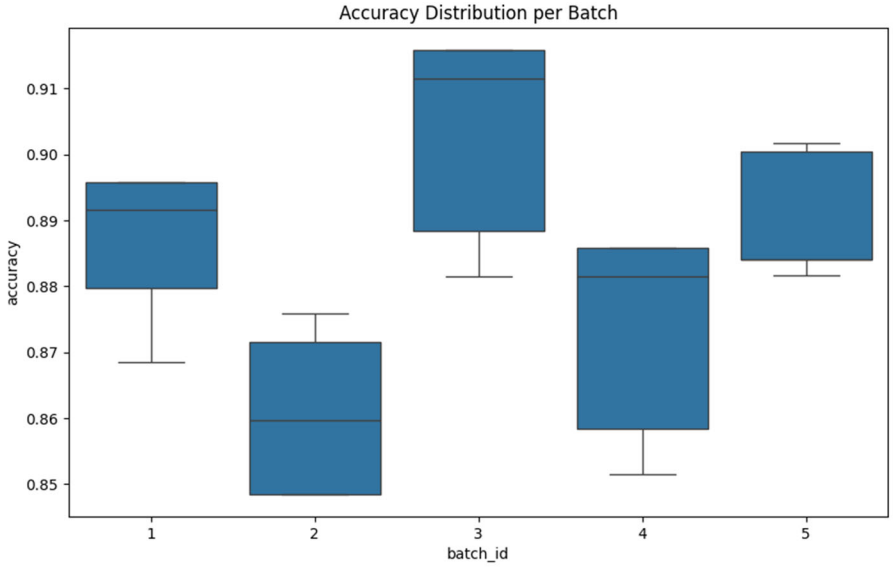


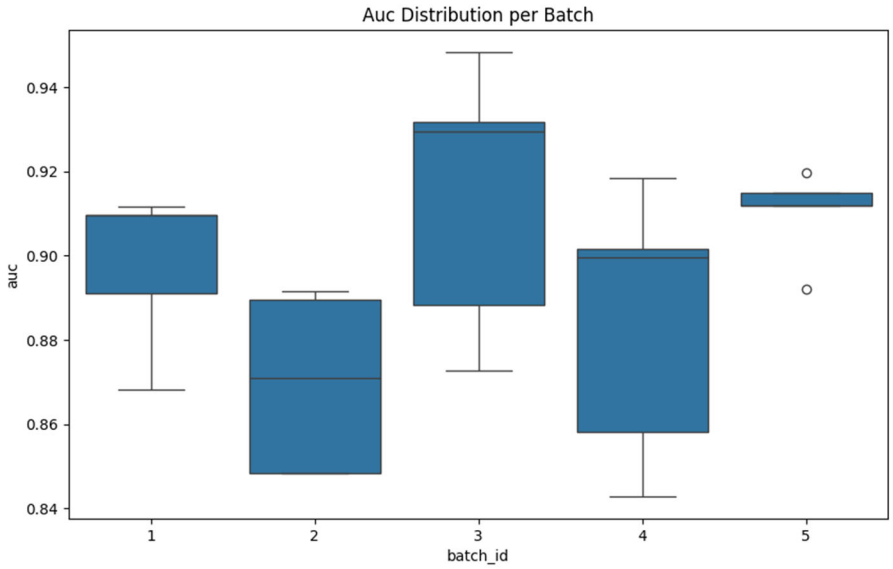

F1_score Mean ± Std per Batch
F1_score
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

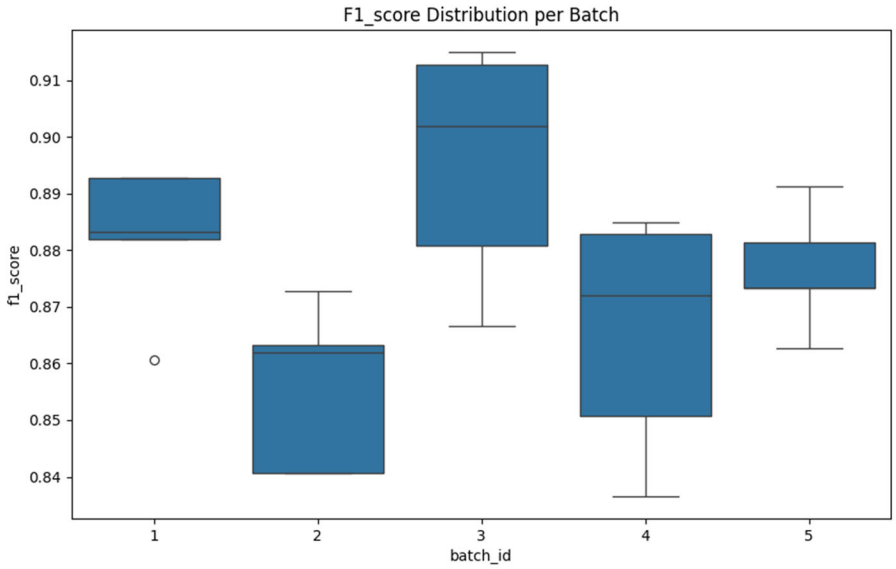

Latency Mean ± Std per Batch
Latency
0
2
4
6
8
10
12
1
2
3
4
5
batch_id

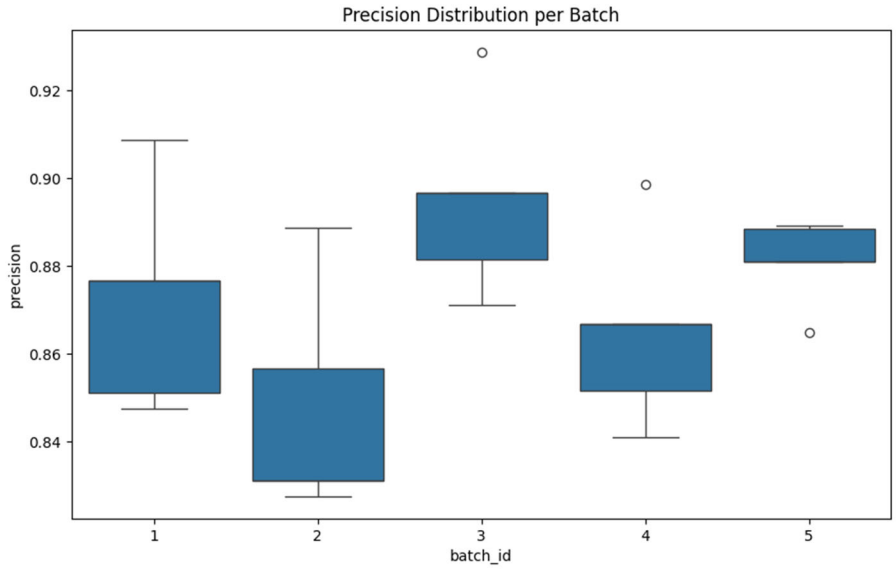

Precision Mean ± Std per Batch
Precision
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

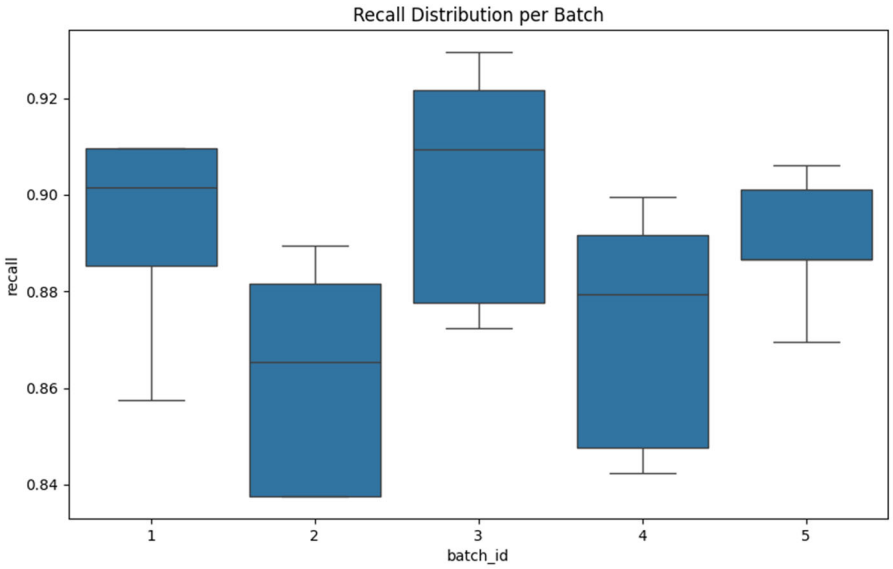

Recall Mean ± Std per Batch
Recall
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

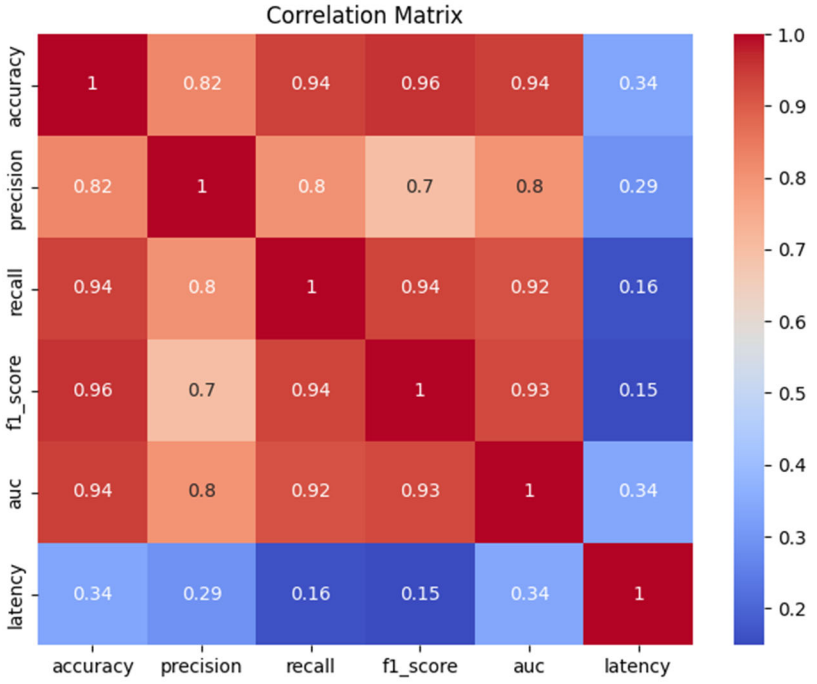

Accuracy Distribution per Batch
accuracy
0.91
0.90
0.89
0.88
0.87
0.86
0.85
1
2
3
4
5
batch_id

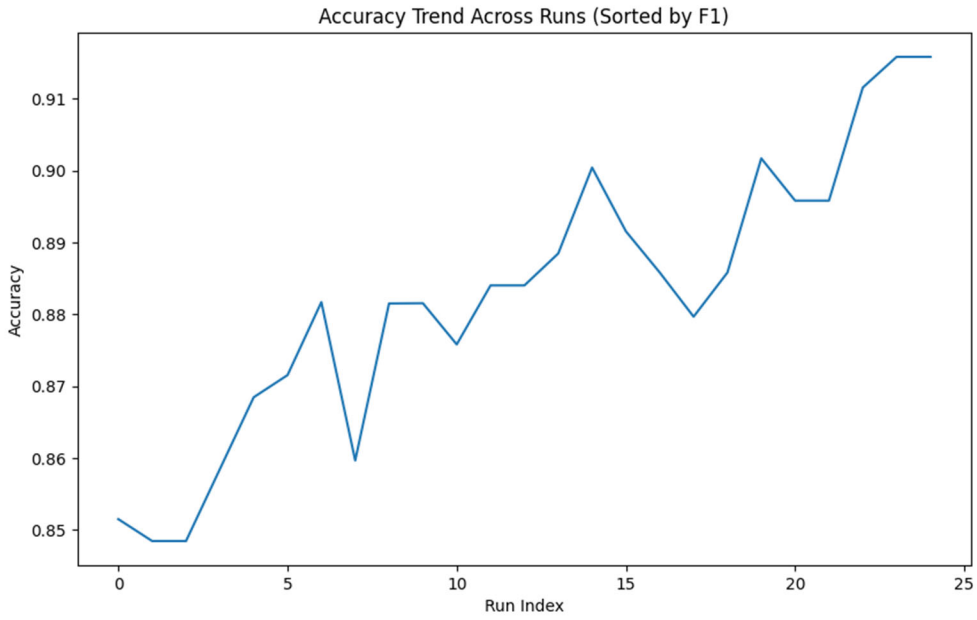

Auc Distribution per Batch
auc
0.94
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5
batch_id

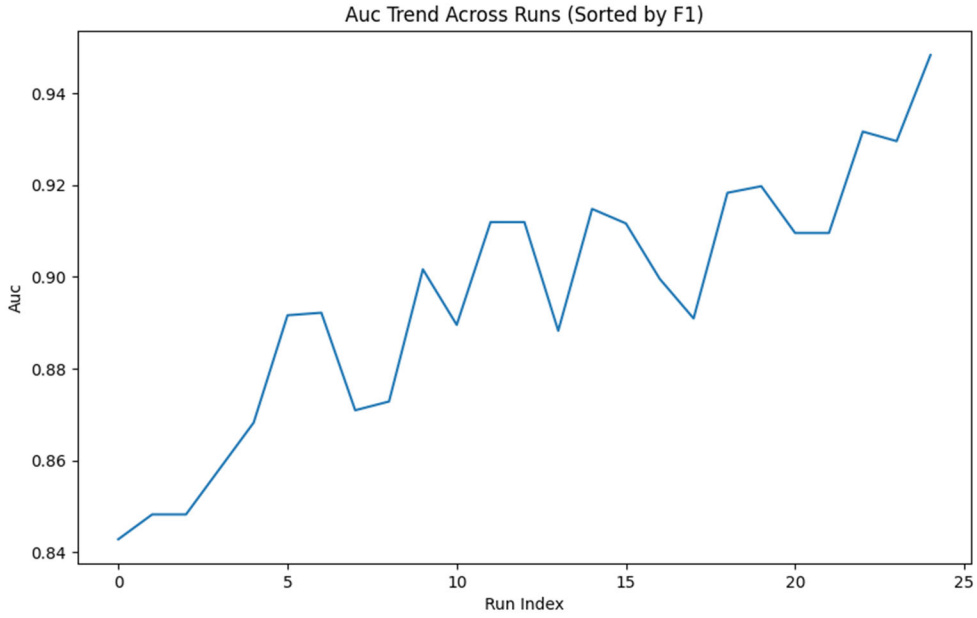

F1_score Distribution per Batch
f1_score
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.84
1
2
3
4
5
batch_id

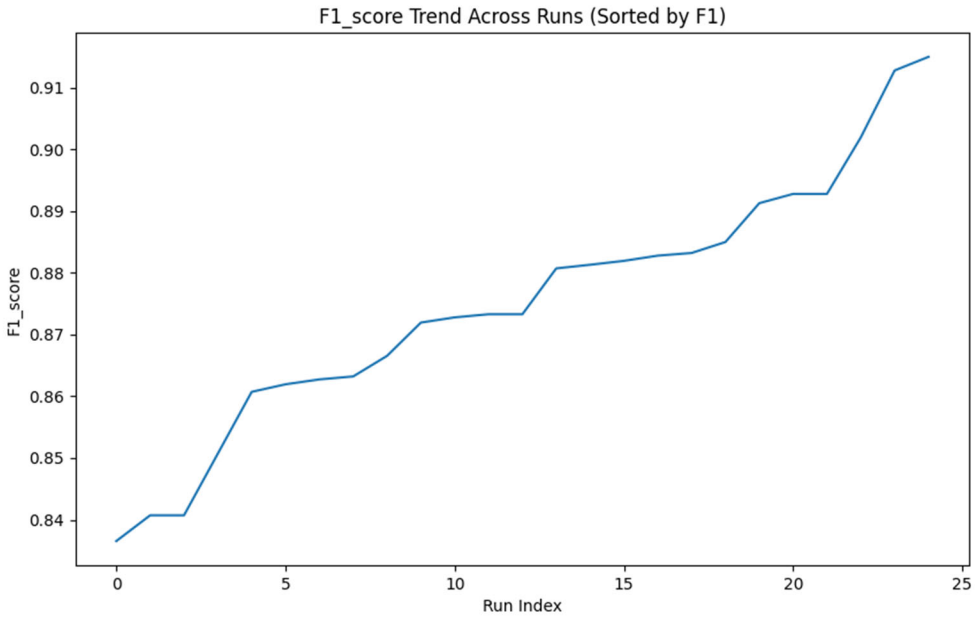

Precision Distribution per Batch
precision
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5
batch_id

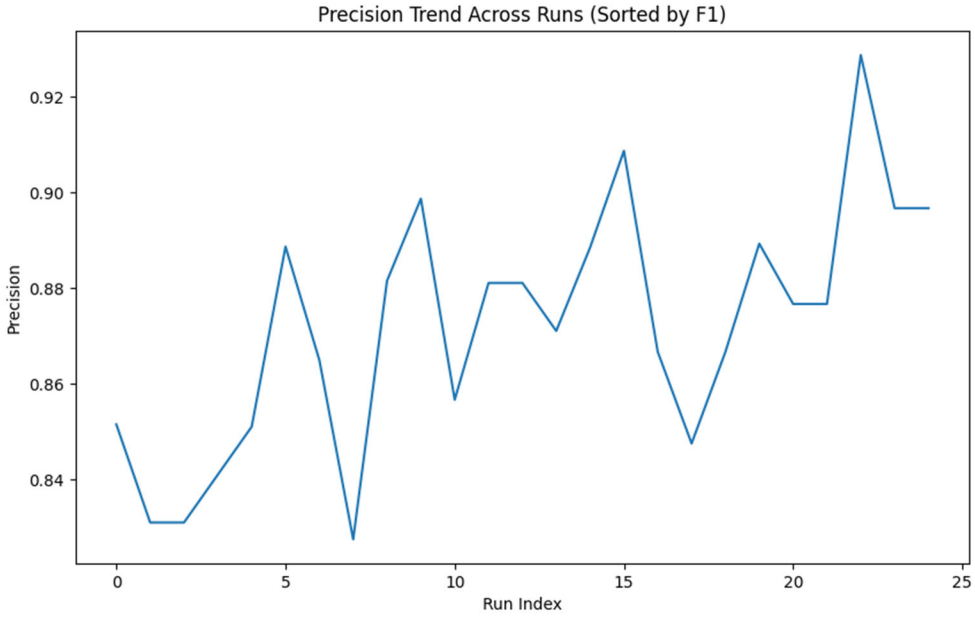

Recall Distribution per Batch
recall
batch_id
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5

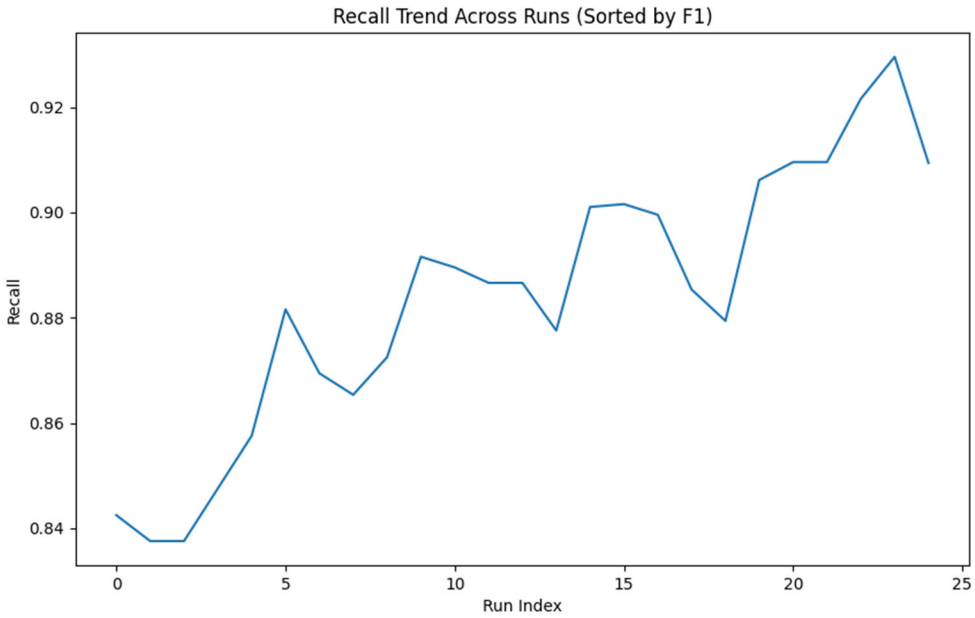

Correlation Matrix
accuracy
precision
recall
f1_score
auc
latency
1 0.82 0.94 0.96 0.94 0.34
0.82 1 0.8 0.7 0.8 0.29
0.94 0.8 1 0.94 0.92 0.16
0.96 0.7 0.94 1 0.93 0.15
0.94 0.8 0.92 0.93 1 0.34
0.34 0.29 0.16 0.15 0.34 1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2


Accuracy Trend Across Runs (Sorted by F1)
Accuracy
Run Index
0.85
0.86
0.87
0.88
0.89
0.90
0.91
0
5
10
15
20
25


Auc Trend Across Runs (Sorted by F1)
Auc
Run Index
0.84
0.86
0.88
0.90
0.92
0.94
0
5
10
15
20
25


F1_score Trend Across Runs (Sorted by F1)
F1_score
Run Index
0.84
0.85
0.86
0.87
0.88
0.89
0.90
0.91
0
5
10
15
20
25


Precision Trend Across Runs (Sorted by F1)
Precision
Run Index
0.84
0.86
0.88
0.90
0.92
0
5
10
15
20
25


Recall Trend Across Runs (Sorted by F1)
Recall
Run Index
0.92
0.90
0.88
0.86
0.84
0
5
10
15
20
25

6. Key Insights & Analysis

- Best Performing Metric: LATENCY with mean 9.411680
- Most Stable Metric: ACCURACY with std 0.018973
- Highest Variance: LATENCY with variance 2.129293
- Best Run: Batch 3 Run run_4 with F1 0.914991
- Worst Run: Batch 4 Run run_5 with F1 0.836546
- Total evaluation samples: 150 across 5 batches